

T0-Theory: Cosmology

Contents

.1	Introduction	2
	Cosmology within the Framework of the T0-Theory	2
	Connection to the T0 Document Series	2
.2	Time-Energy Duality and the Static Universe	3
	Heisenberg's Uncertainty Principle as a Cosmological Principle	3
	Consequences for Standard Cosmology	3
.3	The Cosmic Microwave Background Radiation (CMB)	3
	CMB as ξ -Field Manifestation	3
	CMB Energy Density and Characteristic Length Scale	4
.4	Casimir Effect and ξ -Field Connection	4
	Casimir-CMB Ratio at the Characteristic Length Scale	4
	ξ -Field as Universal Vacuum	5
.5	Cosmic Redshift: Alternative Interpretations	5
	The Mathematical Model of the T0-Theory	5
	Alternative Physical Interpretations	5
	Experimental Distinction of Interpretations	6
	Common Predictions of All Interpretations	7
	Strategic Significance of Multiple Interpretations	7
.6	Structure Formation in the Static ξ -Universe	7
	Continuous Structure Development	7
	ξ -Supported Continuous Creation	7
	Solution to Structure Formation Problems	8
.7	Dimensionless ξ -Hierarchy	8
	Energy Scale Ratios	8
.8	Experimental Predictions and Tests	9
	Precision Casimir Measurements	9
	Electromagnetic ξ -Resonance	9
	Cosmic Tests of Wavelength-Dependent Redshift	9
.9	Cosmological Problems Compared	9
	Comparison: Λ CDM vs. T0 Model	9
	Parameter Count	10
.10	Cosmic Timescales and ξ -Evolution	10
	Characteristic Timescales	10
	Cosmic ξ -Cycles	11
.11	Connection to Dark Matter and Dark Energy	11
	ξ -Field as Dark Matter Alternative	11
	No Dark Energy Required	11
.12	Cosmic Verification through the CMB_En.py Script	11
	Automated Calculations	11
	Reproducible Science	12
.13	Philosophical Implications	12

The World View of T0-Cosmology	12
Epistemological Significance	12
Technological Applications	13

Abstract

This document presents the cosmological aspects of the T0-Theory with the universal ξ -parameter as the foundation for a static, eternally existing universe. Starting from the time-energy duality, it is argued that a classical Big Bang singularity is excluded and replaced by a tiny but finite core with minimal length scale L_0 (from ξ), and it is proposed that the cosmic microwave background radiation (CMB) as well as the Casimir effect be understood as two manifestations of the same ξ -field. As the sixth document of the T0 series, it carries the basic principles of the preceding documents over to cosmological questions.

Note on versions. This English version of the document is more extensive than the German one. Earlier revisions shortened or changed parts of the German version. Where the two differ, this more extensive English version applies.

Status markers. Statements marked **[S]** are *speculative*: a hint or conjecture that has not yet been derived or numerically checked in the corpus. The present corpus deliberately sets the cosmic sector (H_0 , Λ , dark energy, origin of the CMB) aside, because the Standard Model does not derive it either (P39); the cosmological interpretations in this document should be read in that sense.

1 Introduction

Cosmology within the Framework of the T0-Theory

The T0-Theory introduces a relationship between the microscopic quantum vacuum and macroscopic cosmic structures. The cosmological quantities treated in this document are traced back to the parameter $\xi = \frac{4}{3} \times 10^{-4}$.

Key Result

Central Thesis of T0-Cosmology [S]:

The universe is static and eternally existing. The observed cosmic phenomena are interpreted here as manifestations of the ξ -field, not as a consequence of spacetime expansion (cosmic sector, cf. P39).

Connection to the T0 Document Series

This cosmological analysis builds on the fundamental insights of the previous T0 documents:

- **003_T0_Grundlagen_v1_En.pdf:** Geometric parameter ξ and fractal spacetime structure
- **011_T0_Feinstruktur_En.pdf:** Electromagnetic interactions in the ξ -field
- **012_T0_Gravitationskonstante_En.pdf:** Gravitation theory from ξ -geometry
- **006_T0_Teilchenmassen_En.pdf:** Mass spectrum as the basis for cosmic structure formation
- **007_T0_Neutrinos_En.pdf:** Neutrino oscillations in cosmic dimensions

.2 Time-Energy Duality and the Static Universe

Heisenberg's Uncertainty Principle as a Cosmological Principle

Key statement

Starting argument of this document:

According to the argument presented here, Heisenberg's uncertainty principle $\Delta E \times \Delta t \geq \frac{\hbar}{2}$ excludes a classical Big Bang singularity with infinite density; its place is taken by a tiny but finite core with minimal length scale L_0 (from ξ). The corpus keeps this conclusion; the justification via the uncertainty principle, however, has been replaced by the corpus's own time-energy reciprocity $T \cdot E = 1$ (Doc. 306).

In natural units ($\hbar = c = k_B = 1$), the time-energy uncertainty relation reads:

$$\Delta E \times \Delta t \geq \frac{1}{2} \quad (1)$$

In the original version of this document, the chain of argument reads:

- A temporal beginning (Big Bang) would imply $\Delta t = \text{finite}$
- This leads to $\Delta E \rightarrow \infty$ - physically inconsistent
- Therefore, the universe must have existed eternally: $\Delta t = \infty$
- The universe is static, without expanding space

Consequences for Standard Cosmology

Warning

Problems of Big Bang Cosmology as Seen in This Document:

1. **Conflict with Quantum Mechanics:** Finite Δt requires infinite energy (argument of this document; justification replaced by Doc. 306)
2. **Fine-Tuning Problems:** Over 20 free parameters required
3. **Dark Matter/Energy:** 95% unknown components
4. **Hubble Tension:** 9% discrepancy between local and cosmic measurements
5. **Age Problem:** Objects older than the stated age of the universe

.3 The Cosmic Microwave Background Radiation (CMB)

CMB as ξ -Field Manifestation

[S] If there is no Big Bang, the CMB needs an origin other than the $z=1100$ decoupling of standard cosmology. The T0-Theory proposes to explain the CMB through ξ -field quantum fluctuations (cosmic sector, cf. P39).

T0-CMB-Temperature Relation:

$$\frac{T_{\text{CMB}}}{E_\xi} = \frac{16}{9} \xi^2 \quad (2)$$

With $E_\xi = \frac{1}{\xi} = \frac{3}{4} \times 10^4$ (natural units) and $\xi = \frac{4}{3} \times 10^{-4}$, the result is:

$$T_{\text{CMB}} = \frac{16}{9} \xi^2 \times E_\xi \quad (3)$$

$$= \frac{16}{9} \times \left(\frac{4}{3} \times 10^{-4}\right)^2 \times \frac{3}{4} \times 10^4 \quad (4)$$

$$= \frac{16}{9} \times 1.78 \times 10^{-8} \times 7500 \quad (5)$$

$$= 2.37 \times 10^{-4} \text{ (natural units)} \quad (6)$$

Conversion to SI Units: $T_{\text{CMB}} \approx 2.75 \text{ K}$ (measured value 2.725 K)

This is close to the Planck measurement; the dimensionless ratios differ by about 1 % (3.13 versus 3.16×10^{-8} , see the table of dimensionless ξ -ratios).

CMB Energy Density and Characteristic Length Scale

The CMB energy density defines a fundamental characteristic length scale of the ξ -field:

$$\rho_{\text{CMB}} = \frac{\xi}{\ell_\xi^4} \quad (7)$$

From this follows the characteristic ξ -length scale:

$$\ell_\xi = \left(\frac{\xi}{\rho_{\text{CMB}}} \right)^{1/4} \quad (8)$$

Key Result

Characteristic ξ -Length Scale:

Using the experimental CMB data, the result is:

$$\ell_\xi = 100 \text{ } \mu\text{m} \quad (9)$$

This length scale marks the transition region between microscopic quantum effects and macroscopic cosmic phenomena.

.4 Casimir Effect and ξ -Field Connection

Casimir-CMB Ratio at the Characteristic Length Scale

The ratio between Casimir energy density and CMB energy density is compatible with the characteristic ξ -length scale; at $d = \ell_\xi$, however, it is not an independent comparison (see below).

The Casimir energy density at plate separation $d = \ell_\xi$ is:

$$|\rho_{\text{Casimir}}| = \frac{\pi^2 \hbar c}{720 \times \ell_\xi^4} \quad (10)$$

The theoretical ratio yields:

$$\frac{|\rho_{\text{Casimir}}|}{\rho_{\text{CMB}}} = \frac{\pi^2}{720\xi} = \frac{\pi^2 \times 10^4}{960} \approx 102.8 \quad (11)$$

The factor 720 belongs to the energy density between the plates; $\pi^2 \hbar c / (240 d^4)$ is the Casimir pressure. Since ℓ_ξ is fixed from ρ_{CMB} via $\rho_{\text{CMB}} = \xi \hbar c / \ell_\xi^4$, the ratio at $d = \ell_\xi$ is an identity and not an independent comparison.

Experiment

Numerical Comparison:

The Python verification script `CMB_En.py` (available on GitHub:) yields:

- Theoretical value: 102.8 (with $\ell_\xi = 100.24 \mu\text{m}$ from $\rho_{\text{CMB}} = 4.175 \times 10^{-14} \text{ J/m}^3$)
- The same expression in SI units with rounded $\ell_\xi = 100 \mu\text{m}$: 103.8 (not a measured value)
- Deviation: 1.0 % (rounding of ℓ_ξ)

ξ -Field as Universal Vacuum

Key statement

Interpretation [S]:

According to this interpretation, the ξ -field manifests itself both in the free CMB radiation and in the geometrically confined Casimir vacuum. The document reads this as a hint that the ξ -field acts as the universal quantum vacuum; at $d = \ell_\xi$ the numerical comparison is an identity and not independent evidence.

The characteristic ξ -length scale ℓ_ξ is the point where CMB vacuum energy density and Casimir energy density reach comparable orders of magnitude:

$$\text{Free Vacuum: } \rho_{\text{CMB}} = \frac{\pi^2}{15} T_{\text{CMB}}^4 = 2.0 \times 10^{-15} \text{ eV}^4 \text{ (natural units)} \quad (12)$$

$$\text{Confined Vacuum: } |\rho_{\text{Casimir}}| = \frac{\pi^2}{720 d^4} \quad (13)$$

.5 Cosmic Redshift: Alternative Interpretations

The Mathematical Model of the T0-Theory

The T0-Theory provides a mathematical model for the observed cosmic redshift that *allows alternative interpretations*, without committing to a specific physical cause.

Fundamental T0-Redshift Model:

$$z \approx d \cdot C \cdot \xi \quad (\text{see Document 026}) \quad (14)$$

where d is the distance and C the constant from Doc. 026. Since ξ is dimensionless and d a length, $\xi \cdot d$ would not be dimensionless without such a length scale, not even for $\hbar = c = 1$.

Alternative Physical Interpretations

The same mathematical model can be realized through different physical mechanisms:

Interpretation 1: Energy Loss Mechanism

Photons lose energy through interaction with the omnipresent ξ -field:

$$\frac{dE}{dx} = -\frac{\xi E^2}{E_\xi} \quad (15)$$

Physical Assumptions:

- Direct energy transfer from the photon to the ξ -field
- Continuous process over cosmic distances
- No space expansion required

Interpretation 2: Gravitational Deflection by Mass

The redshift arises from cumulative gravitational deflection effects along the light path:

$$z(d) = \int_0^d \xi \cdot \rho_{\text{Matter}}(x) dx \quad (\text{see Document 026}) \quad (16)$$

Physical Assumptions:

- Matter distribution determined by ξ -parameter
- Gravitational frequency shift accumulates over distance
- Static universe with homogeneous matter distribution

Interpretation 3: Spacetime Geometry Effects

The ξ -field structure of spacetime modifies light propagation:

$$ds^2 = \left(1 + \frac{\xi \lambda_0}{E_\xi}\right) dt^2 - dx^2 \quad (17)$$

Physical Assumptions:

- Wavelength-dependent metric coefficients
- ξ -field as fundamental spacetime component
- Geometric cause of frequency shift

Experimental Distinction of Interpretations**Experiment****Tests to Distinguish Mechanisms:****1. Polarization Analysis:**

- Energy Loss: No polarization effects
- Gravitational Deflection: Weak polarization rotation
- Geometric Effects: Specific polarization patterns

2. Temporal Variation:

- Energy Loss: Constant effect
- Gravitational Deflection: Varies with local matter density

- Geometric Effects: Dependent on ξ -field fluctuations
- 3. Spectral Signatures:**
- Energy Loss: Smooth wavelength-dependent curve
 - Gravitational Deflection: Discrete peaks at mass concentrations
 - Geometric Effects: Interference patterns at characteristic frequencies

Common Predictions of All Interpretations

Regardless of the specific mechanism, the T0 model predicts:

Key Result

Universal T0-Redshift Predictions [S] (original state; cf. K6 in Doc. 190):

- **Wavelength Dependence:** $z \propto \lambda_0$
- **Distance Dependence:** $z \propto d$ (linear, not exponential)
- **Characteristic Scale:** Effects maximal at $\lambda \sim \ell_\xi$
- **Ratio of Different Wavelengths:** $z_1/z_2 = \lambda_1/\lambda_2$

Strategic Significance of Multiple Interpretations

Warning

Methodological Advantage:

By offering multiple interpretations, the T0-Theory avoids:

- Premature commitment to a specific mechanism
- Exclusion of experimentally equivalent explanations
- Ideological preferences over physical evidence
- Limitation of future theoretical developments

This corresponds to the principle of scientific objectivity and falsifiability.

.6 Structure Formation in the Static ξ -Universe

Continuous Structure Development

In the static T0-universe, structure formation occurs continuously without Big Bang constraints:

$$\frac{d\rho}{dt} = -\nabla \cdot (\rho \mathbf{v}) + S_\xi(\rho, T, \xi) \quad (18)$$

where S_ξ is the ξ -field source term for continuous matter/energy transformation.

ξ -Supported Continuous Creation

The ξ -field enables continuous matter/energy transformation:

$$\text{Quantum Vacuum} \xrightarrow{\xi} \text{Virtual Particles} \quad (19)$$

$$\text{Virtual Particles} \xrightarrow{\xi^2} \text{Real Particles} \quad (20)$$

$$\text{Real Particles} \xrightarrow{\xi^3} \text{Atomic Nuclei} \quad (21)$$

$$\text{Atomic Nuclei} \xrightarrow{\text{Time}} \text{Stars, Galaxies} \quad (22)$$

The energy balance is maintained by:

$$\rho_{\text{total}} = \rho_{\text{Matter}} + \rho_{\xi\text{-Field}} = \text{constant} \quad (23)$$

Solution to Structure Formation Problems

Key Result

Advantages of T0 Structure Formation [S]:

- **Unlimited Time:** Structures can become arbitrarily old
- **No Fine-Tuning:** Continuous evolution instead of critical initial conditions
- **Hierarchical Development:** From quantum fluctuations to galaxy clusters
- **Stability:** Static universe prevents cosmic catastrophes

.7 Dimensionless ξ -Hierarchy

Energy Scale Ratios

The ξ -relations can be written as dimensionless ratios:

Table 1: Dimensionless ξ -Ratios in Cosmology

Ratio	Expression	Value
CMB Temperature	$\frac{T_{\text{CMB}}}{E_{\xi}}$	3.13×10^{-8}
Theory	$\frac{16}{9} \xi^2$	3.16×10^{-8}
Characteristic Length	$\frac{\ell_{\xi}}{\ell_{\xi}}$	$\xi^{-1/4}$
Casimir-CMB	$\frac{ \rho_{\text{Casimir}} }{\rho_{\text{CMB}}}$	$\frac{\pi^2 \times 10^4}{960}$
Hubble Substitute	$\frac{\xi x}{E_{\xi} \lambda}$	dimensionless
Structure Scale	$\frac{L_{\text{Structure}}}{\ell_{\xi}}$	$(\text{Age}/\tau_{\xi})^{1/4}$

Warning

Structure of the T0-Cosmology Relations:

The theoretical ξ -relations contain only simple numbers:

- Fractions: $\frac{4}{3}, \frac{3}{4}, \frac{16}{9}$
- Powers of Ten: $10^{-4}, 10^3, 10^4$

- Mathematical Constants: π^2

No fitted decimal numbers appear on the theory side; the expressions follow from ξ and integer structural choices that enter as motivated settings.

.8 Experimental Predictions and Tests

Precision Casimir Measurements

Experiment

Casimir Measurement at Characteristic Length Scale:

Casimir force measurements at $d = 100 \mu\text{m}$ should show the theoretical ratio 102.8:1 to the CMB energy density. This number follows from the definition of ℓ_ξ and the standard Casimir formula; a deviation would test the Casimir formula, not the ξ -field.

Experimental Accessibility: $\ell_\xi = 100 \mu\text{m}$ is within the measurable range of modern Casimir experiments.

Electromagnetic ξ -Resonance

Maximum ξ -field-photon coupling at characteristic frequency:

$$\nu_\xi = \frac{c}{\ell_\xi} = \frac{3 \times 10^8}{10^{-4}} = 3 \times 10^{12} \text{ Hz} = 3 \text{ THz} \quad (24)$$

At this frequency, electromagnetic anomalies should occur, measurable with high-precision THz spectrometers.

Cosmic Tests of Wavelength-Dependent Redshift

Note: According to the present state of the corpus, the redshift is achromatic (Doc. 190, K6). The wavelength-dependent tests below reflect the original state of this document.

Experiment

Multi-Wavelength Astronomy:

1. **Galaxy Spectra:** Comparison of UV, optical, and radio redshifts
 2. **Quasar Observations:** Wavelength dependence at high z values
 3. **Gamma-Ray Bursts:** Extreme UV redshift vs. radio components
- The T0-Theory predicts specific ratios that deviate from standard cosmology.

.9 Cosmological Problems Compared

Comparison: Λ CDM vs. T0 Model

The right-hand column gives the interpretation in the static T0 picture; like the whole cosmic sector, it is speculative **[S]** (P39).

Table 2: Cosmological Problems: Standard vs. T0

Problem	Λ CDM	T0 Interpretation
Horizon Problem	Inflation required	Infinite causal connectivity
Flatness Problem	Fine-tuning	Geometry stabilized over infinite time
Monopole Problem	Topological defects	Defects dissipate over infinite time
Lithium Problem	Nucleosynthesis discrepancy	Nucleosynthesis over unlimited time
Age Problem	Objects older than universe	Objects can be arbitrarily old
H_0 Tension	9% discrepancy	No H_0 in static universe
Dark Energy	69% of energy density	Not required
Dark Matter	26% of energy density	ξ -field effects

Parameter Count

Key statement

From 25+ Parameters to a Single One:

- Standard Model of Particle Physics: 19+ parameters
- Λ CDM Cosmology: 6 parameters
- **T0-Theory: 1 Parameter (ξ)**, plus integer structural choices as motivated settings and SI anchors for unit conversion

This corresponds to a reduction of the continuous parameters by 96%. According to the present state of the corpus, however, the cosmological sector requires its own, independently set scale (P39, Doc. 306).

.10 Cosmic Timescales and ξ -Evolution

Characteristic Timescales

The ξ -field defines fundamental timescales for cosmic processes:

$$\tau_\xi = \frac{\ell_\xi}{c} = \frac{10^{-4}}{3 \times 10^8} = 3.3 \times 10^{-13} \text{ s} \quad (25)$$

Longer timescales arise from ξ -hierarchies:

$$\tau_{\text{Atom}} = \frac{\tau_{\xi}}{\xi^2} \approx 10^{-5} \text{ s} \quad (26)$$

$$\tau_{\text{Molecule}} = \frac{\tau_{\xi}}{\xi^3} \approx 0.14 \text{ s} \quad (27)$$

$$\tau_{\text{Cell}} = \frac{\tau_{\xi}}{\xi^4} \approx 1.1 \times 10^3 \text{ s} \approx 18 \text{ min} \quad (28)$$

Cosmic ξ -Cycles

In this picture, the static T0-universe undergoes ξ -driven cycles [S]:

1. **Matter Accumulation:** ξ -field $\rightarrow$ particles $\rightarrow$ structures
2. **Structure Maturity:** Galaxies, stars, planets
3. **Energy Return:** Hawking radiation $\rightarrow \xi$ -field
4. **Cycle Restart:** New matter generation

.11 Connection to Dark Matter and Dark Energy

ξ -Field as Dark Matter Alternative

Key Result

ξ -Field as an Interpretation of Dark Matter [S]:

- Gravitationally acting through energy-momentum tensor
- Electromagnetically neutral (detectable only via specific resonances)
- Correct cosmological energy density at $\Delta m \sim \xi \times m_{\text{Planck}}$
- Intended to explain galaxy rotation curves without new particles (not worked out quantitatively here)

No Dark Energy Required

In the static T0-universe, no dark energy is required [S]:

- No accelerated expansion to explain
- Supernova observations explainable by wavelength-dependent redshift (original state; cf. K6 in Doc. 190)
- CMB anisotropies arise from ξ -field fluctuations, not primordial density perturbations

.12 Cosmic Verification through the CMB_En.py Script

Automated Calculations

The Python verification script `CMB_En.py` (available on GitHub:) performs calculations of the T0-cosmological relations named here:

- **Characteristic ξ -Length Scale:** $\ell_{\xi} = 100 \mu\text{m}$
- **CMB-Temperature Verification:** Theoretical vs. experimental

- **Casimir-CMB Ratio:** 102.8 (or 103.8 with rounded $\ell_\xi = 100 \mu\text{m}$; 1.0 % deviation from rounding, not a comparison with a measurement)
- **Scaling Behavior:** Tested over 5 orders of magnitude
- **Energy Density Consistency:** Dimensional analysis of the energy densities

Experiment

Automated Verification of T0-Cosmology:

The script generates:

- Detailed log files with all calculation steps
- Markdown reports for scientific documentation
- LaTeX documents for publications
- JSON data export for further analyses

Result: The computed quantities lie close to the comparison values (Casimir-CMB ratio: 1.0 % deviation, solely from rounding ℓ_ξ).

Reproducible Science

The automation of T0 calculations supports:

- **Transparency:** All calculation steps documented
- **Reproducibility:** Identical results on every run
- **Scalability:** Easy extension for new tests
- **Validation:** Automatic consistency checks

.13 Philosophical Implications

The World View of T0-Cosmology

Key statement

Interpretation of T0-Cosmology [S]:

The document interprets the universe as the expression of a mathematical order that is described at its core by the parameter ξ .

Possible philosophical conclusions from this interpretation:

- **Eternal Existence:** The universe had no beginning and will have no end
- **Mathematical Order:** All structures follow exact geometric principles
- **Universal Unity:** Quantum and cosmic scales are fundamentally connected
- **Deterministic Evolution:** Randomness is excluded at the fundamental level

Epistemological Significance

The T0-Theory stands for the view that:

- Complex phenomena can be derived from simple principles
- Mathematical simplicity is a useful heuristic criterion
- Reductionism to a fundamental parameter is possible

- The universe is rationally comprehensible

Technological Applications

Conceivable, purely speculative technological consequences **[S]**:

- **ξ -Field Manipulation:** Control over fundamental vacuum properties
- **Energy Extraction:** Tapping into the cosmic ξ -field
- **Communication:** ξ -based instantaneous information transfer
- **Transport:** ξ -field-supported propulsion systems

Bibliography

- [1] Pascher, J. (2025). *T0-Theory: Fundamental Principles*. T0 Document Series, Document 1.
- [2] Pascher, J. (2025). *T0-Theory: Gravitational Constant*. T0 Document Series, Document 3.
- [3] Pascher, J. (2025). *T0-Theory: Particle Masses*. T0 Document Series, Document 4.
- [4] Pascher, J. (2025). *T0-Model Casimir-CMB Verification Script*. GitHub Repository.
- [5] Pascher, J. (2025). *T0-Theory: Cosmic Relations*. Project Documentation.
- [6] Heisenberg, W. (1927). *On the Perceptual Content of Quantum Theoretical Kinematics and Mechanics*. Zeitschrift für Physik, 43(3-4), 172–198.
- [7] Planck Collaboration (2020). *Planck 2018 results. VI. Cosmological parameters*. Astronomy & Astrophysics, 641, A6.
- [8] Casimir, H. B. G. (1948). *On the attraction between two perfectly conducting plates*. Proceedings of the Royal Netherlands Academy of Arts and Sciences, 51(7), 793–795.
- [9] Lamoreaux, S. K. (1997). *Demonstration of the Casimir force in the 0.6 to 6 μm range*. Physical Review Letters, 78(1), 5–8.
- [10] Riess, A. G., et al. (2022). *A Comprehensive Measurement of the Local Value of the Hubble Constant*. The Astrophysical Journal Letters, 934(1), L7.
- [11] Weinberg, S. (1989). *The cosmological constant problem*. Reviews of Modern Physics, 61(1), 1–23.
- [12] Peebles, P. J. E. (2003). *The Lambda-Cold Dark Matter cosmological model*. Proceedings of the National Academy of Sciences, 100(8), 4421–4426.
- [13] Einstein, A. (1917). *Cosmological Considerations on the General Theory of Relativity*. Sitzungsberichte der Königlich Preußischen Akademie der Wissenschaften, 142–152.
- [14] Hubble, E. (1929). *A relation between distance and radial velocity among extragalactic nebulae*. Proceedings of the National Academy of Sciences, 15(3), 168–173.
- [15] Friedmann, A. (1922). *On the Curvature of Space*. Zeitschrift für Physik, 10(1), 377–386.